

- 1) Create dir mern-demo
- 2) Create two sub-dir in this dir
 - a. mern-frontend
 - b. mern-backend
- 3) Open mern-demo in vscode
- 4) Open terminal , go to frontend dir
 - a. give command `npx create-react-app .`
- 5) Open mern-backend in new terminal
- 6) Give command
 - a. Npm install init
 - b. Give command `init -y`
 - c. Give command `npm install express mongoose cors`
- 7) The project structure of backend and front is ready now
- 8) Create a file server.js in mern-backend
- 9) Copy the code provided in server.js
- 10) Now go to mern-frontend
- 11) Copy the App.js code provided in this file
- 12) Change App.css
- 13) Start MongoDB
 - a. Create connection BCATRG
 - b. Create database studentdb
 - c. Create collection students
 - d. Import csv format data, with fields rollno, name, course, semester
- 14) Now goto mern backend to start the DBserver
 - a. Give command `node server.js`
- 15) Now go to mern-frontend
 - a. Give command `npm start`

See the student list in browser

Server.js

```
const express = require('express');  
const mongoose = require('mongoose')  
const cors = require('cors');
```

```
const app = express();  
app.use(cors());
```

```
// Connect to MongoDB  
mongoose.connect('mongodb://localhost:27017/studentdb')  
  .then(() => console.log('MongoDB Connected'))  
  .catch(err => console.error(err));
```

```
// Define Student schema  
const studentSchema = new mongoose.Schema({  
  rollno: Number,  
  name: String,  
  course: String,  
  semester: Number  
});
```

```
const Student = mongoose.model('Student',  
studentSchema,'students');
```

```
// Sample route: get all students  
app.get('/students', async (req, res) => {  
  const students = await Student.find();  
  res.json(students);  
});  
app.listen(5000, () => {  
  console.log('Server running on port 5000');  
});
```

App.js

```
import React, { useEffect, useState } from 'react';
```

```
import './App.css';
```

```
function App() {
```

```
  const [students, setStudents] = useState([]);
```

```
  useEffect(() => {
```

```
    fetch('http://localhost:5000/students')
```

```
      .then(res => res.json())
```

```
      .then(data => setStudents(data))
```

```
      .catch(err => console.error(err));
```

```
  }, []);
```

```
  return (
```

```
    <div>
```

```
      <h1>Student List</h1>
```

```
      <table border="1" cellPadding="10" cellSpacing="0">
```

```
        <thead>
```

```
          <tr>
```

```
            <th>Roll No</th>
```

```
            <th>Name</th>
```

```
            <th>Course</th>
```

```
            <th>Semester</th>
```

```
          </tr>
```

```
        </thead>
```

```
        <tbody>
```

```
          {students.map((s) => (
```

```
            <tr key={s.rollno}>
```

```
              <td>{s.rollno}</td>
```

```
              <td>{s.name}</td>
```

```
              <td>{s.course}</td>
```

```
        <td>{s.semester}</td>
      </tr>
    )}
  </tbody>
</table>
</div>
);
}
export default App;
```

App.css

```
.App {  
  text-align: center;  
}
```

```
.App-logo {  
  height: 40vmin;  
  pointer-events: none;  
}
```

```
@media (prefers-reduced-motion: no-preference) {  
  .App-logo {  
    animation: App-logo-spin infinite 20s linear;  
  }  
}
```

```
.App-header {  
  background-color: #282c34;  
  min-height: 100vh;  
  display: flex;  
  flex-direction: column;  
  align-items: center;  
  justify-content: center;  
  font-size: calc(10px + 2vmin);  
  color: white;  
}
```

```
.App-link {  
  color: #61dafb;  
}
```

```
@keyframes App-logo-spin {
```

```
from {  
  transform: rotate(0deg);  
}  
to {  
  transform: rotate(360deg);  
}  
}
```

```
/* Full page */  
body {  
  margin: 0;  
  font-family: 'Segoe UI', sans-serif;  
  background: #f4f7fc;  
}
```

```
/* Center content */  
.container {  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  min-height: 100vh;  
}
```

```
/* Card */  
.table-card {  
  width: 80%;  
  max-width: 900px;  
  background: white;  
  padding: 30px;  
  border-radius: 12px;  
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.1);  
}
```

```
/* Heading */  
h1 {  
    text-align: center;  
    color: #2c3e50;  
    margin-bottom: 25px;  
}
```

```
/* Table */  
table {  
    width: 100%;  
    border-collapse: collapse;  
    overflow: hidden;  
    border-radius: 10px;  
}
```

```
/* Header */  
thead {  
    background: #3498db;  
    color: white;  
}
```

```
th {  
    padding: 15px;  
    text-align: left;  
}
```

```
/* Body */  
td {  
    padding: 12px 15px;  
    border-bottom: 1px solid #ddd;  
}
```

```
/* Zebra stripes */  
tbody tr:nth-child(even) {  
    background: #f8fafc;  
}
```

```
/* Hover effect */  
tbody tr:hover {  
    background: #d6eaf8;  
    transition: 0.3s;  
}
```